

Sentiment Analysis on Product Reviews

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Submission Date: 3 May 2026

Introduction

This project analyzes customer reviews from an e-commerce dataset to determine sentiment (Positive, Negative, Neutral). The goal is to automate understanding of customer satisfaction.

Methodology

The Amazon Fine Food Reviews dataset was used (first 5000 rows). Data was cleaned by removing null and duplicate values. Sentiment analysis was performed using TextBlob, classifying polarity into Positive (>0), Negative (<0), and Neutral ($=0$).

Results & Analysis

Most reviews were positive, indicating high customer satisfaction. Negative reviews highlighted product quality and packaging issues. Word frequency analysis showed common complaint terms.

Key Insights

- Around 70–80% reviews are positive
- Negative reviews mention poor quality and packaging
- Some low ratings had neutral/positive wording
- Sentiment and ratings don't always match

Recommendation

Businesses should improve packaging quality and maintain product consistency to reduce negative feedback.

Conclusion

Sentiment analysis helps extract insights from large text data and supports better business decisions.